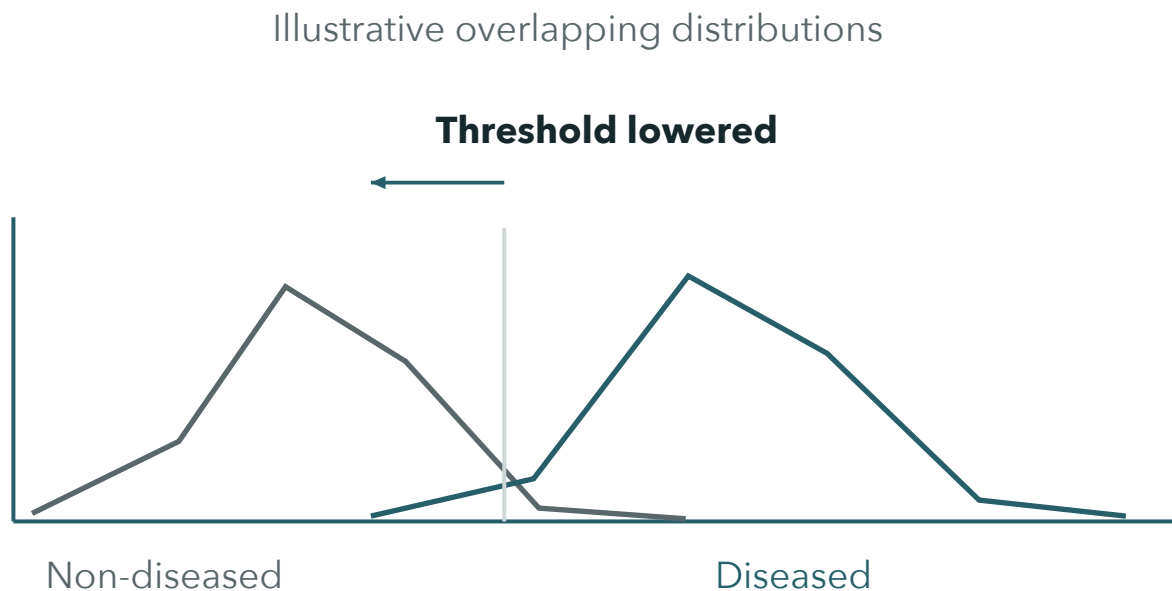


Diagnostic thresholds and study biases

Part 1 | A threshold selects an operating point

A continuous biomarker becomes a binary test only after a positive-negative boundary is chosen.



For a positive-above test, lowering the threshold includes additional diseased and non-diseased observations.

Read the direction

With the same participants and reference labels, lowering the threshold generally increases sensitivity and decreases specificity. The test's biological measurement has not changed; its classification rule has. A threshold cannot be judged without considering the consequence of false positives and false negatives.

Predictive value depends on the population

Transporting a test to a lower-prevalence setting changes the interpretation of a positive result.

Held fixed

Feature	Quantity	Changes with prevalence?
Sensitivity	Positive among diseased	Not in this stipulated comparison
Specificity	Negative among non-diseased	Not in this stipulated comparison
PPV	Diseased among positive	Yes

Original comparison

A test is used in a referral clinic and a low-risk screening population with identical sensitivity and specificity. The latter positive results contain a larger proportion of false positives. This is a probability consequence of the changed prior distribution, not proof that the instrument became technically less accurate.

Verification can depend on the test result

Selective reference testing changes the sample used to estimate accuracy.

Partial verification pathway

- Apply the index test to all
- ↓
- Refer mostly index-positive patients
- ↓
- Reference-standard labels remain missing for many negatives
- ↓
- Analyse an unrepresentative verified subgroup

Bias distinction

This differs from incorporation bias, in which the index result contributes to the reference diagnosis. It also differs from spectrum bias due to an unrepresentative disease mix. Identify the selection mechanism actually described rather than calling every diagnostic design defect observer bias.

AUC does not choose the clinical threshold

A summary across operating points can conceal the part of the curve that matters clinically.

Report

Feature	Useful information	Limit
ROC curve	Sensitivity versus false-positive rate	Threshold still needs selection
AUC	Overall ranking discrimination	Not a specific treatment rule
Chosen operating point	Errors at relevant threshold	Consequences depend on setting

Ophthalmic detail

If both eyes contribute observations, correlation can also affect uncertainty in ROC estimates and comparisons. A valid ranking summary and an invalid standard error can coexist. Address the unit of analysis as well as the accuracy metric.

A valid comparison needs a credible reference

Every accuracy calculation inherits the definition and acquisition of the disease label.

Check

Feature	Question	Risk if ignored
Reference	Independent and appropriate?	Circular classification
Verification	Applied to relevant participants?	Selective ascertainment
Blinding	Reader knows index result?	Interpretation influenced

Examination approach

Find the bias that follows directly from the stated method. A plausible problem that was never described is weaker than a demonstrated one. Then state which missing data or design repair would address it, rather than merely naming the bias.